

PCB DESIGNING FOR ENGINEERS

With the announcement of government of India's manufacturing policy expected soon, the stage seems to be well set for robust growth in the PCB market in near future, opening up a minefield of opportunities for PCB designers



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For any electronics product, printed circuit board (PCB) forms the basic foundation for interconnecting and packaging. PCBs are used to mechanically support and electrically connect electronic components using conductive pathways, tracks or signal traces etched from copper sheets laminated on a non-conductive

substrate (source: Wikipedia). PCBs are also referred to as printed wiring boards (PWBs), or etched wiring boards, which have evolved over the years from uncomplicated single- and double-sided plated-through-hole (PTH) to become multi-layered PCBs.

The fortunes of the PCB industry depend on the health of IT, telecom, automobile, electronic gaming and consumer electronics. Due to the rapid

growth of the electronics industry worldwide, the global PCB market has witnessed a remarkable growth over the last two decades.

What's the market like?

Japan and Asia-Pacific collectively contribute a major share to the global PCB market. Amongst Asian countries, India and China are central to the growth of the PCB industry, thanks to the rapid expansion of electronic manufacturing bases. This growth is triggered by the shifting of production activities and facilities of key multinational players in these countries.

The demand from automotive electronics segment drives a positive outlook for the PCB market over the next few years. As electronic components account for a fair portion of the overall value of materials consumed in cars' production, the stage seems to be well set for robust growth in the PCB market in near future. "Without PCB, we can not expect any electronic circuit. Since a PCB provides physical support as well as interconnections between different electronic components, there exist great opportunities as well as very good growth in this field worldwide," says Shavinder Singla, deputy engineer in PCB Lab at Centre for Development of Advanced Computing (CDAC), Mohali.

K. Krithiga, technical lead at CADD Center, adds, "For electricals and electronics engineering (EEE) and electronics communication engineering (EC)